TASK 3

# Deep-Dive Analysis & Interactive Dashboarding

Data Analytics Internship • KPI Framework, Example Analysis & Dashboard Blueprint

**Purpose**
Define measurable KPIs, analyze customer behavior using a sample e-commerce scenario, and specify an interactive dashboard that supports business decisions.

## 1. Business Problem & Scope

Business question: How can an online store improve conversion and repeat purchasing while protecting customer value? The analysis below uses **illustrative sample figures** for demonstration. Replace them with actual project data before submitting as evidence of real analysis.

## 2. Core KPI Definitions

| KPI | Formula | Why it matters |
|---|---|---|
| Conversion Rate | Orders ÷ Sessions × 100 | Measures how effectively visits become orders. |
| Customer Churn Rate | Customers lost during period ÷ Customers at start × 100 | Highlights retention risk. |
| Average Order Value (AOV) | Revenue ÷ Number of orders | Shows average revenue per order. |
| Repeat Purchase Rate | Customers with 2+ orders ÷ Customers × 100 | Tracks customer loyalty. |
| Cart Abandonment Rate | (Carts created – Orders) ÷ Carts created × 100 | Identifies checkout friction. |

## 3. Deep-Dive Analysis (Illustrative)

### A. Funnel analysis

Example funnel: 10,000 sessions → 2,500 product views → 900 add-to-carts → 540 checkout starts → 360 completed orders. Session-to-order conversion is 3.6%. From checkout starts to completed orders, completion is 66.7%, suggesting checkout is a useful area to investigate.

### B. Cohort / retention analysis

Compare customers by their first-purchase month and measure the share who return in later months. If the newest cohort shows lower repeat purchasing, investigate acquisition source, product mix, delivery experience, and post-purchase communication before concluding that retention has worsened.

### C. Customer segmentation

Create practical segments such as New, Repeat, High-Value, and At-Risk customers using order count, recency, and spend. Example rules: New = one order in the last 30 days; Repeat = at least two orders; High-Value = top 20% by spend; At-Risk = no order in 90 days. Adjust thresholds to the real business.

## 4. Interactive Dashboard Blueprint

Recommended tool: Power BI, Tableau, or Looker Studio. Add date, channel, region, and customer-segment filters. The overview page should show KPI cards, a monthly trend line, a conversion funnel, retention by cohort, and

segment-level revenue. Selecting a segment should filter the other visuals.

### Dashboard quality checklist

• Use clear titles, units, and date ranges.
• Include tooltips and cross-filtering.
• Show comparisons against the previous period or target.
• Avoid misleading truncated axes and unnecessary decoration.
• Validate totals against the source dataset.

## 5. Findings & Recommended Actions

1. Investigate the checkout drop-off with device, payment method, and error-rate breakdowns.
2. Compare repeat-purchase rates by acquisition channel and first-purchase month.
3. Test targeted win-back messages for at-risk customers and track incremental lift.
4. Review KPI definitions with stakeholders before publishing.

## Deliverables checklist

■ KPI definitions and formulas documented
■ Deep-dive report completed with validated data
■ Interactive dashboard published and link checked
■ Short walkthrough video recorded

**Important:** This PDF is a structured sample report; it does not itself create or host a live dashboard. Add your own dashboard URL and replace illustrative numbers with actual results.